MILITARY SPECIFICATION

# SCRAPER, 14-18 CUBIC YARD MINIMUM STRUCK CAPACITY, OPEN BOWL, SINGLE-DIESEL-ENGINE-DRIVEN, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE.

1.1 Scope. This specification covers a commercial open bowl, pneumatic-tired, two-axle, 14-18 cubic yard minimum struck capacity, single-diesel-engine-driven, articulated frame steer scraper.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

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| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-L-2104 | - Lubricating Oil, Internal Combustion. |
| MIL-L-2105 | - Lubricating Oil, Gear, Multipurpose. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3805

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| MIL-T-3351 | - Tractor, Full-Tracked, Low Speed; Tractor, Wheeled, Agricultural; and Tractor, Wheeled, Industrial; and Their Attachments, Packaging of. |
| MIL-M-7866 | - Molybdenum Disulfide, Technical, Lubrication Grade. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-I-14511 | - Insulation Sheet, Cellular, Plastic: Thermal. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-S-40626 | - Sign Kit, Vehicle Class. |
| MIL-A-46153 | - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package. |
| MIL-L-46167 | - Lubricating Oil, Internal Combustion Engine, Arctic. |
| MIL-C-46168 | - Coating, Aliphatic Polyurethane, Chemical Agent Resistant. |
| MIL-B-46176 | - Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative. |

STANDARDS

MILITARY

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| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-209 | - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |
| MIL-STD-1410 | - Method for Selection of Industrial Engines for End Item Application. |
| MIL-STD-1472 | - Human Engineering Design Criteria for Military Systems, Equipment and Facilities. |
| MIL-STD-1474 | - Noise Limits for Army Materiel. |
| MS35424 | - Light, Marker, Clearance - Blackout, 24 Volt. |
| MS51318 | - Headlight, Blackout, 24 Volt, Waterproof. |
| MS51330 | - Stop Light - Tail-light, Vehicular 24 Volt, Blackout Tail, Blackout Stop. |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.



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TECHNICAL INFORMATION PACKAGE (TIP)

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HKE, Fort Belvoir VA 22060.)

DEPARTMENT OF TRANSPORTATION (DOT)

DOT Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 General. The scraper shall be the latest model of the standard product of the contractor and shall have demonstrated industry acceptability by having been manufactured, marketed, and sold in significant numbers to the commercial public for the same or similar application, interfacing with the same or similar components, except as specified herein, for at least 1 year prior to invitation for bids or request for proposal. Product improvements incorporated in the commercial production units during this 1 year period are acceptable. When specified requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment, shall be furnished. The scraper shall be equipped with optional equipment as specified herein. Optional equipment is defined as equipment not standard with the scraper but which has been furnished to the commercial customer, such as special features or allied equipment. The scraper shall be equipped with all components necessary to enable the scraper to function reliably and efficiently in sustained operation. The scraper shall conform in quality to that normally



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provided to industrial and commercial customers. The scraper shall conform to all Federal laws and regulations governing safety, noise levels, and pollution, which are in effect for this type of equipment at the time of manufacture.

3.2 Description. The scraper shall be a commercial open bowl model, 14-18 cubic yard minimum struck capacity, with one diesel engine, four pneumatic tires, two axles, two single driving front wheels, two single non-driving rear wheels, and articulated steering.

3.3 First article (preproduction models). The contractor shall furnish one scraper for examination and demonstration within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce scrapers that comply with the requirements of this specification. The examination and demonstrations shall be as specified in Section 4. The examination and all demonstrations conducted by the contractor shall be subject to surveillance and approval by the Government (see 6.3). Acceptance of the preproduction models will be dependent on successful completion of the examination, contractor demonstrations, and the Government testing. At the option of the contractor, two scrapers may be furnished to expedite examination and demonstration.

3.4 Production scrapers. All scrapers furnished under this specification shall be physically and mechanically identical to the first article scraper approved by the Government, except that product improvements incorporated in the contractor's standard commercial product lines are permitted at no cost to the Government, subject to the following:

(a) Physical, mechanical, and performance characteristics have not been degraded.

(b) New components involving Government selected repair parts are directly interchangeable with the superseded component or part.

(c) The Government is notified of all changes to any item in the recommended parts list where fit and function is effected a minimum of 45 days prior to the production date of the change. When the changes and alterations have been incorporated in the production scraper, the Government shall be furnished the production effectivity by serial number and dates.

(d) Appropriate changes are made to the Government validated operating, maintenance and parts manuals before delivery.

(e) The contractor shall be responsible to provide repair parts for the service life of the scraper.

3.5 Material. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).



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3.6 Physical characteristics.

3.6.1 Weight limit. The operating weight of the scraper shall be between 58,000 pounds and 68,000 pounds. Operating weight shall be determined with the ROPS/FOPS, Cab, all standard and required accessories installed; and coolant, lubricating, hydraulic, and fuel systems filled to capacity.

3.7 Bowl. The bowl shall have a minimum struck capacity of between 14 and 18 cubic yards when measured in accordance with SAE J741.

3.7.1 Cutting and spread depth. The bowl shall be capable of power down and shall have a minimum cut depth of 12 inches. The spread depth shall be at least 18 inches.

3.7.2 Cut width. The width of the scraper cut shall be at least 116 inches.

3.7.3 Cutting edge. The cutting edges shall be replaceable and reversible. If two or more cutting edges are the same size they shall be interchangeable.

3.7.4 Dumping of the bowl. The contractor's standard apron and ejector shall be operable and furnished in the same combination and configuration as supplied to commercial industry. In whatever combination and configuration, they shall provide for complete dumping of the bowl.

3.7.5 Ground clearance. The ground clearance of the tractor or scraper when fully loaded shall be at least 18 inches.

3.8 Safety and human factors.

3.8.1 Safety. All available standard options and accessories shall be included to provide protection of maintenance and operating personnel from rotating or reciprocating parts and from parts that are subject to high operational temperatures and so located as to be a hazard. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the scraper can be set into motion. Exhaust discharges shall be directed so as not to endanger personnel or obstruct operators view. All platforms and steps shall have antiskid surfaces. A decal shall be furnished and shall be located in the area of the articulation point warning of personnel clearance during frame articulation and placement of the steering frame lock for scraper transport and as required to provide for safe maintenance. Contractor's standard grab irons shall be provided.

3.8.2 Sound level. The scraper shall be equipped with all available standard and optional sound suppression equipment. If the sound level in the area occupied by the operator exceeds category D of MIL-STD-1474 when measured in accordance with SAE J919, the contractor shall provide a caution plate and appropriate warning in the technical manuals in accordance with MIL-STD-1474. If the sound level with cab door(s) open is 85 dB(A) or greater, when it is less than 85 dB(A) with the doors closed, a caution plate shall be provided stating:

"CAUTION, HEARING PROTECTION REQUIRED
WHILE OPERATING WITH DOORS OPEN."



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The caution plate shall have a yellow background with black lettering and shall be made of a corrosion resistant material. The caution plate shall be permanently affixed to the scraper in a conspicuous, protected location in the operator's compartment. The exterior sound level of the scraper shall not exceed 88 dB(A) when measured in accordance with SAE-J88.

3.8.3 Human factors. The characteristics of the scraper with ROPS cab shall allow for operation and maintenance by the 5th to 95th percentile soldier in accordance with MIL-STD-1472, SAE J833, and SAE J925 to the extent commercially available. Operating controls shall be located within the maximum control space in accordance with SAE J898 and MIL-STD-1472 to the extent commercially available.

3.8.4 Rollover protective structure (ROPS) and falling objects protective structure (FOPS) with enclosed cab. The scraper shall be equipped with an enclosed insulated and sound suppressed cab integral with the ROPS/FOPS structure. The ROPS shall conform to SAE J1040 and the FOPS shall conform to SAE J231. The sound suppressed enclosed cab shall consist of: a front safety glass windshield with windshield wiper, rear window, right side window, and left side door for the operator to enter and exit the operator's compartment. The door shall have window panes. The left side door shall be able to open and be held open by an integral part of the scraper during all types of operations. Except for the door holding device, the enclosed cab shall be the contractor's cab normally furnished to commercial industry. Air conditioning is not required. Interior and exterior rear-view mirrors shall be provided.

3.9 Performance. The scraper shall withstand the usage encountered in military operations such as being push-loaded, limited self-loading, hauling, dumping and spreading soil, crossing rough terrain, fording streams (see 3.9.4), and negotiating steep longitudinal and sidehill slopes (see 3.9.3), without damage. The scraper shall be capable of driving at maximum travel speed in transport, being towed by another vehicle, and limited towing of other equipment.

3.9.1 Environmental conditions. Without the use of any external means, equipped with only standard and optional equipment, the engine shall start within 5 minutes and the scraper shall operate under full-load conditions within 15 minutes after engine start in any ambient temperature from minus 25° F to plus 120° F. Cold start aids must be integral with the scraper and no aids may be used which rely upon external support such as an electrical power source. Hand-held aerosol cans are not permitted. The scraper shall be capable of operating in hot humid tropical environments and dry dusty environments. The scraper shall be capable of being stored for extended periods in temperatures ranging from -30° F to +160° F without damage.

3.9.2 Travel speed. The scraper with full heaped load shall be capable of a forward travel speed of not less than 30 mph on level pavement.

3.9.3 Stability. The scraper shall negotiate 35 percent side slopes empty, 20 percent side slopes loaded, and descend 40 percent longitudinal slopes loaded without any wheel of the scraper losing contact with the ground. The engine lubricating and fuel supply systems shall maintain operating pressure under these conditions.



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3.9.4 Fording. The scraper shall start and operate during and after fording operations in fresh water or salt water to any depth up to thirty inches including wave action without damage and without leakage of water into reservoirs containing lubricants.

3.9.5 Construction tasks. The scraper shall be capable of performing: Being push-loaded, limited self-loading, hauling, dumping and spreading soil.

3.10 Diesel engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The engine shall be the heavy-duty diesel furnished commercially including all systems, components, accessories, and auxiliaries. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow. The engine shall be the same engine that is used in the manufacturer's standard commercial scraper. An air cleaner restriction gage shall be furnished. The engine lubricating and fuel supply systems shall maintain operating pressure on 35 percent side slopes and 40 percent longitudinal slopes.

3.10.1 Engine horsepower. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816 shall not be less than 325 horsepower at the governed rpm.

3.10.2 Engine cooling system. The contractor's standard or optional cooling system meeting the requirements of 3.9 thru 3.9.4 shall be provided.

3.10.3 Fuel tank. The fuel tank shall be of a type regularly furnished as standard equipment with capacity for a minimum of 10 hours of continuous normal operation. The fuel tank shall be equipped with a means for draining water and sediment from the bottom of the tank. Fuel capacity shall be verified during conduction of construction task 4.5.2.5. Label(s) shall be placed near the fill port to indicate "Diesel Fuel" fill port.

3.10.4 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. Sampling valve shall be labeled crankcase or transmission, whichever is appropriate to indicate the type of oil sampled.

3.10.5 Engine Enclosure. The engine shall be enclosed and covered with the contractor's standard or optional engine hood, engine side panels with quick disconnect latches, or engine side doors which are horizontally or vertically hinged.



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3.10.6 Tachometer drive. A heavy duty key type tachometer drive shall be provided and shall have a 5/32-inch female connection in accordance with SAE J678.

3.11 Drive train. A drive train consisting of a torque converter, full powershift transmission, front drive axle with differential and planetary final drives, and all necessary connecting components shall be furnished. A hydraulic or electric retarder shall be furnished. The transmission shall provide not less than five forward speeds and one reverse speed. A locking or limited slip differential shall be furnished in the drive axle.

3.12 Wheels and tires. Commercial standard wheel and rim combinations shall be provided. The tires shall be traction type (E2) tubeless, and shall have a tread pattern suitable for scraper operation. The size of the tires shall be 29.5 by 29 with a 28 ply rating.

3.13 Operator's compartment.

3.13.1 Seat. The contractor's standard commercial operator's seat to include air or hydraulic suspension shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, visibility of the work zone, and all instrument and indicators mounted in the operator's compartment, when seated with seat belt fastened and properly adjusted.

3.13.2 Seat belt. The contractor's standard commercial seat belt conforming to SAE J386 shall be properly installed.

3.13.3 Instruments, indicators, and gauges. The following instruments, indicators, or gauges shall be provided:

(a) Engine oil pressure gauge.
(b) Engine coolant temperature gauge.
(c) Transmission/torque converter oil temperature gauge.
(d) Ammeter or voltmeter.
(e) Engine hourmeter or service meter.
(f) Air pressure gauge (when equipped with air system).
(g) Low air pressure warning buzzer (when equipped with air system).
(h) Fuel level indicator (gauge or dipstick).
(i) Hydraulic oil level indicator (dipstick, sight tube, or gauge).
(j) Air cleaner restriction gauge.
(k) Tachometer.

Additional instruments, indicators, and gauges standard with the contractor's commercial practice which contribute to the safety and reliability of the scraper operation shall be provided.

3.13.4 Cab heater/defroster. A heater for the area occupied by the operator shall be provided. A front windshield and rear window(s) defroster shall be provided and may be integral with the heater.

3.14 Transportability. The scraper shall be capable of being transported by truck trailer, rail, ocean-going carriers, and C5A aircraft. Cargo envelopes



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for the C5A are shown in figure 1. The scraper shall withstand any forces (impact) encountered in all modes of transportation without damage, permanent deformation, breakage of any structural member or component, or loss of ability of the scraper to perform all of its intended functions. The scraper shall be equipped with lifting provisions and shall be capable of being lifted without damage using slings (and spreader bars, if needed). The scraper shall be equipped with tiedown provisions for transport by the above modes. Stencil or decal marking shall be applied to the scraper at each lifting and tiedown point. Complete diagrams and instructions for lifting the scraper and tying down the scraper on freight cars, and truck trailers, ocean going vessels, and the C5A aircraft shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport. A steering frame lock in accordance with SAE J276 shall be provided. The contractor shall provide sufficient data to the Government in the Technical Information Package to confirm that the scraper being supplied can be loaded and transported in the C5A aircraft.

3.14.1 Lifting provisions. The scraper shall be provided with four lifting provisions conforming to MIL-STD-209, class 1 or 3, type II. The provisions shall enable the scraper to be lifted in the normal traveling or operating position. Lifting provisions shall be positioned so that the lifting strain shall be in line with longitudinal axis of the eye of the provision. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provision without weld failure or permanent deformation of the scraper or the provisions.

3.14.2 Tiedown provisions. The scraper shall be provided with integral provisions to permit tiedown of the scraper to the floor or deck of the transportation medium. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type II and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the scraper or the provision. Tiedown provisions may also be used as lifting provisions when such provisions meet the requirements specified in 3.14.1.

3.15 Towing devices. Tow hook(s) or other suitable means of attaching a tow cable or tow bar shall be provided on the front and rear of the scraper.

3.16 Pusher block. The contractor's standard rear pusher-block shall be provided to permit the scraper to be push-loaded.

3.17 Electrical system. The manufacturer's standard or optional electrical system shall be provided which meets the requirements specified herein to include the blackout lighting system. A 24-volt starting system shall be furnished.

3.17.1 Battery. The battery or batteries shall be the contractor's standard or optional batteries for cold weather starting as specified in 3.9.1.

3.17.2 Alternator. The rated output of the alternator shall be not less than 50 amps.



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3.17.3 Lighting. The following lights with associated circuits, circuit breakers, or other protective devices, and switches shall be provided.

(a) Two headlights.
(b) Two front working lights (flood).
(c) Two combination rear stop and taillights.
(d) Rear floodlight(s) to illuminate cutting edge.
(e) Instrument panel lighting.
(f) Blackout lights in accordance with 3.17.5.

Additional lights or lighting standard with the contractor's commercial practice shall be provided.

3.17.4 Electromagnetic radiation. The scraper shall comply with the electromagnetic radiation limits of SAE J551. Suppressors, where used shall conform to SAE J552.

3.17.5 Blackout lighting. A blackout lighting system shall be furnished to meet the following minimum requirements:

(a) Each vehicle will be equipped with the following blackout lights:

(1) One left front corner light conforming to MS51318.
(2) Two front corner marker lights conforming to MS35424-1.
(3) Two rear corner marker and stop/taillights conforming to MS51330.

(b) The blackout system shall be controlled by an interior switch, readily accessible to the driver, which will prevent accidental disengagement of the blackout system from the blackout mode.
(c) All regular exterior lights and horn, if electric, shall be made inoperable automatically when in the blackout mode.
(d) The interior lighting shall have a variable brightness control switch readily accessible to the driver.
(e) A separate wiring harness shall be provided for the blackout lights only if the required blackout lights cannot be powered using the standard wiring harness.

3.18 Horn. A conventional operator controlled horn shall be furnished.

3.19 Back-up alarm. A back-up alarm conforming to SAE J994, type C or type D shall be furnished. The back-up alarm signal shall be audible above and distinguishable from the vehicle noise while operating.

3.20 Hydraulic system. The scraper shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all scraper components. Label(s) shall be placed near the fill port to indicate type of fill port and recommended military lubricants to be used.



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3.21 Steering system. The scraper shall have an articulated frame type steering system which shall provide full control of the scraper under all normal operating conditions. The scraper shall be equipped with an emergency steering system conforming to SAE J53 that will give the operator steering control in the event of engine failure. Label(s) shall be placed near the fill port to indicate type of fill port and recommended military lubricants to be used.

3.22 Braking system. The scraper shall be equipped with power assisted brakes on all four wheels. A parking brake and an emergency braking system shall be provided. The braking system shall conform to SAE J1152. When nonpetroleum based hydraulic brake fluid is used in a braking system, it shall be the silicone type in accordance with MIL-B-46176. Label(s) shall be placed near the fill port to indicate type of fill port and recommended military lubricants to be used.

3.22.1 Air system. The scraper shall be equipped with the contractor's standard air system to include (as a minimum): compressor, air tank(s), and service outlet.

3.23 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each scraper shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924, and MIL-G-23827 as applicable. Initial servicing and run-in may be accomplished with lubricants standard to the contractor's commercial practice (except for the braking system) if compatible with above military lubricants. Major lubricant points shall have label(s) near them indicating type of lubricant.

3.24 Nonstandard and special tools. Nonstandard and special tools required for equipment operation and operator maintenance of the scraper shall be furnished with each scraper and shall easily fit into the storage compartment or toolbox as described in 3.25.

3.25 Toolbox. A permanent lockable metal storage compartment(s) or toolbox is required. This compartment or toolbox shall be of a sufficient size to hold all tools required in 3.24.

3.26 Vehicle weight classification sign kit. The contractor shall apply the Government furnished vehicle weight classification numbers to each vehicle utilizing a vehicle weight classification number kit conforming to MIL-S-40626. The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the vehicle. Instructions for changing the number based on the vehicle configuration shall be provided in the operator's manual.

3.27 Information plates. All information plates shall be a permanent type, permanently affixed in protected conspicuous positions. Plates shall conform to MIL-P-514 or contractors standard plates or labels subject to the approval of the contracting officer. In addition to the information plates the Government specifies, the contractor shall furnish and affix all other information plates in accordance with the standard commercial practice.



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3.27.1 Identification plates and marking. Each scraper shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, serial number, weight, date of manufacture and date of delivery. All attachments or components removed or disassembled for shipment shall be matchmarked for proper reassembly. The scraper and each disassembled attachment shall be identified in accordance with the contractor's standard method.

3.27.2 Instruction plate. The scraper shall be equipped with instruction plates including warnings and cautions suitably located describing any special or important procedure to be followed in operating and servicing the scraper.

3.27.3 Shipping data plate(s). Shipping data plate(s) for each scraper shall be furnished and shall show the silhouette of the scraper indicating the location and capacity of tiedown provisions, lifting provisions, the center of balance, and shipping weight. The plates shall be attached to the left side of the scraper and shall show the locations as viewed from the left side.

3.28 Treatment and painting. The portions of the scraper normally painted shall be cleaned, treated, and primed in accordance with MIL-T-704, type F and the finishing coat shall be in accordance with MIL-C-46168, color forest green.

3.29 Government-furnished equipment (GFE). The following items shall be furnished by the Government (see 6.4).

(a) Container NSN 7520-00-559-9618 complete with maintenance forms. (The contractor shall furnish the fasteners and install the containers on the scraper.)
(b) Vehicle weight classifications sign kit, NSN 9905-00-565-6267.

3.30 Technical publications. Such technical publications as are specified (see 6.2), shall be furnished.

3.31 Stenciling instructions. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown, lifting, tire pressure, and center of gravity.

3.32 Technical information package. The contractor shall complete and submit a technical information package within the time frame specified (see 6.2). The information shall depict the complete scraper and its components exactly as it will be furnished subject to 3.4 (c). Notation shall be made of all components which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used.

3.33 Basic issue item. When specified (see 6.2), each item in the basic issue item list shall be furnished with each scraper.

3.34 Vehicle classification data sheet. The contractor shall complete and submit the vehicle classification data sheet, figure 2, within the time frame specified (see 6.2).



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3.35 Winterized scraper. When specified (see 6.2), a quantity of scrapers as specified (see 6.2) shall be winterized to permit starting and operation in arctic climates at ambient temperatures to minus 50° F and storage at ambient temperatures to minus 60° F. The contractor shall furnish one scraper for examination and demonstration within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce winterized scrapers that comply with the requirements of this specification.

3.35.1 Winterization kit. The kit shall consist of the components for preheating scraper engine coolant and battery box(s); also insulated sound suppressed ROPS cab, hot water personnel heater (equipped with one or more shut-off valve(s) in the coolant lines to stop and to vary the flow of coolant to the heater), defrosting system for cab windows, front and rear windshield wipers, radiator winterfront, insulated engine and radiator compartments, insulated battery box(s), battery finned connectors, and an alcohol evaporator when air brakes are furnished. Winterization components shall not affect the performance of the scraper and all other scraper requirements shall be met as specified herein.

3.35.2 Heater and heat exchanger. The Eberspacher "Swingfire" heater, Part No. 26-9002, shall be utilized for preheat purposes. When specified (see 6.2), a quantity of Swingfire heaters as specified (see 6.2) shall be furnished. The Eberspacher heat exchanger (water jacket), Part No. 26-9013, compatible with the Swingfire heater shall be provided for heating the engine coolant. The heat exchanger shall be located near the engine compartment for easy accessibility for insertion of heater. The heater unit starting operation shall be provided by the scraper's 24-volt electrical system using a wiring harness with Bosh Receptacle, Part No. 135050012, compatible with the heater starter cable connector. After heater start and insertion into heat exchanger, preheating of the engine coolant shall be provided by thermo-siphon action. The heater exhaust hot air shall be routed to the battery box(s) for hot air drenching evenly of the battery box(s) and then routed away from battery box(s) to the atmosphere. The hot air flow shall not be impeded by excessive back pressure or leakage. To retain the hot air, the system shall be insulated and heat shielded, as required.

3.35.3 Insulation. Insulation shall comply with MIL-I-14511, 1/2 inch. Adhesive for bonding the insulation shall retain its adhesive properties at ambient temperatures to minus 50° F.

3.35.4 Performance. The winterization kit shall be of such design that the following criteria can be met when the scraper has been cold soaked for 24 hours at a temperature of minus 50° F:

(a) Engine start within 5 minutes after preheating engine coolant and battery box(s) for 90 minutes.
(b) Within 1 hour after personnel heater start, the heater shall have raised and stabilized the cab temperature to a desired plus 40° F in the area occupied by the operator's torso, seated.



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(c) Within 1 hour after personnel heater start, the defrosting system shall have cleared sufficient glassed area for safe operation of the scraper and visibility of the scraper work area normally seen when there is no frost on the windows.
(d) All scraper bowl controls shall be operational within 15 minutes after engine start-up.
(e) The transmission and brake controls shall be functional within 15 minutes after engine start-up.

3.35.5 Kit installation. The winterized scraper shall be capable of performing limited self-loading, hauling, dumping and spreading soil, as well as being push-loaded.

3.36 Vandalism protection. The contractor's standard or optional anti-vandalism package shall be furnished with each scraper.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to monitor, observe, or perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection. The first article (preproduction model) scraper (see 6.2), furnished by the contractor shall be examined (see 4.3.1) and demonstrated (see 4.3.2), to determine compliance with the requirements of this specification. Prior to the examination and demonstration of the scraper, the following shall be performed:

(a) Service the scraper with the lubricants specified.
(b) Break-in the scraper as prescribed by the contractor.

4.3.1 Preproduction examination. Prior to the preproduction demonstration, the scraper shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection. Prior to the preproduction demonstration, the scraper shall be examined and compared with this specification, with the contractor's published brochures, with commercial manuals, and with the completed Technical Information Package. Any discrepancy between the specification, brochures, manuals, or Technical Information Package



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and the preproduction models shall be cause for rejection. Prior to the preproduction demonstration, the certifications, and certified test reports required by 4.5.2.10 and 4.5.2.11 shall be examined. Incomplete, inaccurate, or otherwise unacceptable certifications or certified test reports shall be cause for rejection.

TABLE I. Examination schedule.

| Preprod. model(s) | Indiv. | Defect | Requirement paragraph |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| X | X | 101. Scraper description not as specified. | 3.2 |
| X | X | 102. Cutting depth not as specified. | 3.7.1 |
| X | X | 103. Spread depth not as specified. | 3.7.1 |
| X | X | 104. Cut width not as specified. | 3.7.2 |
| X | X | 105. Cutting edge not as specified. | 3.7.3 |
| X | X | 106. Apron and ejector not as specified. | 3.7.4 |
| X | X | 107. Ground clearance not as specified. | 3.7.5 |
| X | X | 108. Safety characteristics not as specified. | 3.8.1 |
| X | X | 109. Articulation warning decal not as specified. | 3.8.1 |
| X | X | 110. Sound level caution sign not as specified. | 3.8.2 |
| X | X | 111. Human factor characteristics not as specified. | 3.8.3 |
| X | X | 112. Cab not as specified. | 3.8.4 |
| X | X | 113. ROPs and FOPs not furnished. | 3.8.4 |
| X | X | 114. Windshield wiper and rear view mirrors not furnished. | 3.8.4 |
| X | X | 115. Engine not as specified. | 3.10 |
| X | X | 116. Engine cooling system not as specified. | 3.10.2 |
| X | X | 117. Fuel tank not as specified. | 3.10.3 |
| X | X | 118. Oil sampling valve not as specified. | 3.10.4 |
| X | X | 119. Engine enclosures not as specified. | 3.10.5 |
| X | X | 120. Tachometer drive not as specified. | 3.10.6 |
| X | X | 121. Drive train not as specified. | 3.11 |
| X | X | 122. Wheels and tires not as specified. | 3.12 |
| X | X | 123. Seat not as specified. | 3.13.1 |
| X | X | 124. Seat belt not as specified. | 3.13.2 |
| X | X | 125. Instruments, indicators, and gauges not as specified. | 3.13.3 |
| X | X | 126. Cab heater not as specified. | 3.13.4 |
| X | X | 127. Transportability not as specified. | 3.14 |
| X | X | 128. Steering frame lock not as specified. | 3.14 |
| X | X | 129. Lifting provision not as specified. | 3.14.1 |
| X | X | 130. Tiedown provisions not as specified. | 3.14.2 |
| X | X | 131. Towing devices not as specified. | 3.15 |
| X | X | 132. Pusher block not as specified. | 3.16 |
| X | X | 133. Electrical system not as specified. | 3.17 |
| X | X | 134. Battery not as specified. | 3.17.1 |
| X | X | 135. Alternator not as specified. | 3.17.2 |
| X | X | 136. Lighting not as specified. | 3.17.3 |



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TABLE I. Examination schedule. (Cont'd)

| Preprod. model(s) | Indiv. | Defect | Requirement paragraph |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| X | X | 137. Blackout lights not as specified. | 3.17.5 |
| X | X | 138. Horn not as specified. | 3.18 |
| X | X | 139. Back up alarm not as specified. | 3.19 |
| X | X | 140. Hydraulic system not as specified. | 3.20 |
| X | X | 141. Steering system not as specified. | 3.21 |
| X | X | 142. Braking system not as specified. | 3.22 |
| X | X | 143. Air system not as specified. | 3.22.1 |
| X | X | 144. Lubricants not as specified. | 3.23 |
| X | X | 145. Nonstandard and special tools not as specified or furnished. | 3.24 |
| X | X | 146. Tool box not as specified. | 3.25 |
| X | X | 147. Vehicle weight classification sign kit not as specified. | 3.26 |
| X | X | 148. Identification plates and marking not as specified. | 3.27.1 |
| X | X | 149. Instruction plates not as specified. | 3.27.2 |
| X | X | 150. Shipping data plates not as specified. | 3.27.3 |
| X | X | 151. Treatment and painting not as specified. | 3.28 |
| X | X | 152. Government-furnished equipment not installed on unit. | 3.29 |
| X | X | 153. Technical publications not as specified. | 3.30 |
| X | X | 154. Stenciling instructions not furnished. | 3.31 |
| X | - | 155. Technical information package not as specified. | 3.32 |
| X | X | 156. Basic issue items (if specified) not furnished. | 3.33 |
| X | - | 157. Vehicle classification data sheet not furnished. | 3.34 |
| X | X | 158. Winterization kit not as specified, when applicable | 3.35.1 |
| X | X | 159. Heater and heat exchanger not as specified, when applicable. | 3.35.2 |
| X | X | 160. Insulation not as specified, when applicable. | 3.35.3 |
| X | X | 161. Vandalism package not as specified. | 3.36 |
| X | X | 162. Parts or components missing, from the vehicle. | Section 3 |
| X | X | 163. Assembly incomplete or incorrect. | Section 3 |
| X | X | 164. Cracked or broken parts or assemblies. | Section 3 |
| X | X | 165. Permanent deformation of any part or assembly. | Section 3 |
| X | X | 166. Leaks in coolant, fuel, oil lines or cylinders. | Section 3 |
| - | X | 167. Material not of same quality as used on preproduction scraper. | Section 3 |
| X | X | 168. Scraper not as specified. | Section 3 |



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4.3.2 Preproduction demonstration. Upon successful completion of the examination specified in 4.3.1, the preproduction model scraper shall be subject to the demonstrations marked "X" in column 1 of table II. Failure of any demonstration or any demonstration reports shall be cause for rejection.

TABLE II. Demonstration schedule.

| Preprod model(s) | Indiv. | Test | Test paragraph | Requirement paragraph |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| X | X | Production run-in. | 4.5.2.1 | Section 3 |
| X | - | Weight. | 4.5.2.2 | 3.6.1, 3.14 |
| X | - | Sound level. | 4.5.2.3 | 3.8.2 |
| X | - | Travel speed. | 4.5.2.4 | 3.9.2 |
| X | - | Construction tasks. | 4.5.2.5 | 3.9.5 |
| X | - | Lifting provisions. | 4.5.2.6 | 3.14.1 |
| X | - | Tiedown provision. | 4.5.2.7 | 3.14.2 |
| X | - | Winterization when applicable. | 4.5.2.8 | 3.35.4 |
| X | - | Kit installation when applicable. | 4.5.2.9 | 3.35.5 |

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the demonstration specified in 4.4.2, each scraper shall be examined for the defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual demonstration. Each scraper shall be demonstrated as specified in column 2 of table II. Failure of any demonstration shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with table I.

4.5.2 Demonstrations. Demonstrations shall be conducted in accordance with table II.

4.5.2.1 Production run-in. Start and operate each scraper as follows: Operate the scraper in all gears forward and reverse at maximum speeds commensurate with conditions for one-half hour. Stop and restart the engine not less than five times. Operate all hydraulic controls for five-minute periods. Hold each hydraulic control in each actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the scraper through not less than 10 full right and 10 full left turns. Make not less than five abrupt panic stops from a speed not less than 20 mph using the service brake. Release and reapply the parking brake not less than five times. Turn on



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all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hydraulic fluid shall be cause for rejection of the scraper by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that scrapers previously accepted are similarly defective unless evidence satisfactory to the contracting officer, is furnished by the contractor that they are not similarly defective.

4.5.2.2 Weight. The weight of the scraper shall be measured with a scale calibrated within 6 months of the demonstration. Inability to meet the minimum and maximum weight limits of 3.6.1 shall constitute failure of this demonstration.

4.5.2.3 Sound level. The contractor shall perform sound level tests in accordance with SAE J88 and SAE J919. Certified prior test data is acceptable if available. Nonconformance to the requirements of 3.8.2 shall constitute failure of this demonstration.

4.5.2.4 Travel speed. The travel speed shall be measured to prove conformance to 3.9.2. Nonconformance to 3.9.2 shall constitute failure of this demonstration.

4.5.2.5 Construction task. The scraper shall be demonstrated as follows for a total of 20 hours to prove conformance to 3.8.1, 3.9, and 3.10.3: self-load (5 hours of cycles), load while being pushed by a dozer (15 hours of cycles), including dumping and spreading undistributed or recompacted soil. The haul and return distances shall be at least 1500 feet. All lighting shall remain on throughout this demonstration. This demonstration shall include at least five stops of the scraper with a full heaped load from a speed of 30 miles per hour to 0 miles per hour using maximum braking without locking the wheels. The scraper fuel capacity shall be measured during conduction of the construction task to prove compliance with the fuel capacity requirements of 3.10.3. Nonconformance to either 3.8.1, 3.9, or 3.10.3 or any malfunction, damage, or breakage of mechanical, hydraulic, or electrical component shall constitute failure of this demonstration.

4.5.2.6 Lifting provisions. The lifting provisions shall be demonstrated to prove conformance to 3.14.1. Nonconformance to 3.14.1, weld failure, or permanent deformation of the scraper or the lifting provisions shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

(a) Lift the scraper and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the scraper. Spreader bars are permitted. Determine the force and angles of application for each provision. Measure the clearance between each sling and the scraper.
(b) Restrain the scraper by anchoring the main frame and subject each slinging provision to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in (a) above. Hold each load for a minimum of 90 seconds.



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4.5.2.7 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.14.2. Apply a force equal to the limit load as specified in MIL-STD-209 to each of the tiedown provisions. Nonconformance to 3.14.2, weld failure or permanent deformation of the scraper or the tiedown provisions shall constitute failure of this demonstration.

4.5.2.8 Winterization. The scraper shall be cold soaked in a temperature controlled chamber to a temperature of minus 50° F and demonstrated as follows to prove conformance to 3.35.4.

(a) Preparation.

1. Insure that the scraper has been serviced with arctic coolant, fuels, lubricants and hydraulic oils.
2. Service Swingfire heater with arctic gasoline.
3. Place scraper in temperature controlled chamber.
4. Instrument the scraper to record as a minimum the following temperatures:

   a. Exhaust, engine heater outlet.
   b. Air, battery box(s), inlet, outlet, and all aides.
   c. Air, operator's compartment, operator torso (seated position).
   d. Air, operator's compartment, floor, roof, sides, (midway to top).
   e. Air, hot water heater outlet.
   f. Electrolyte, all battery(s).
   g. Coolant, engine block inlet.
   h. Coolant, engine block outlet.
   i. Oil, engine sump and gallery.

(b) Procedure.

1. Subject scraper to a stabilized temperature of minus 50° F for 24 hours.
2. Bring Swingfire heater into cold chamber. Connect heater to scraper electrical power source and start heater. (After heater start disconnect from power source).
3. Insert heater into heat exchanger and preheat engine coolant and battery(s) for 90 minutes.
4. Record temperatures continously or at intervals not greater than five minutes.
5. Remove heater from heat exchanger and shut off motor.
6. Start engine from scraper battery(s).
7. Start personnel heater and defrost fans, 30 minutes after engine start.
8. One hour after heater start, record operator compartment air temperatures and check defrost.
9. Operate scraper steering and bowl controls for 10 minutes, within the dimensional limits possible within the test chamber.
10. Engage and operate transmission and brake system as dimensional limits of test chamber permits.

Nonconformance to 3.35.4 shall constitute failure of this demonstration.



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4.5.2.9 Installation test. The purpose of this test is to insure the proper fit, form and function of the winterization kit and that no operational or safety deficiencies exist after installation of the kit on the scraper for which it is intended. The scraper shall be demonstrated for 10 hours to prove conformance to 3.35.5. It shall include limited self loading, being push loaded, hauling, and spreading and dumping soil.

4.5.2.10 Certification. The contractor shall certify in writing to the contracting officer that the scraper complies with the following:

(a) Environmental condition requirements of 3.9.1.
(b) Stability requirement of 3.9.3.
(c) Fording requirement of 3.9.4.
(d) Engine requirements of 3.10.
(e) Horsepower requirements of 3.10.1.
(f) Transportability requirement of 3.14.
(g) Steering system requirements of 3.21.
(h) Braking system requirements of 3.22.

If a certification is not furnished, the requirement shall be demonstrated. Nonconformance to one or more of the above shall constitute failure of this demonstration.

4.5.2.11 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the following:

(a) Bowl capacity requirements of 3.7.
(b) ROPS/FOPS requirement of 3.8.4.
(c) Electromagnetic radiation requirements of 3.17.4.

If certified test reports are not available, these requirements shall be demonstrated. Nonconformance shall constitute failure of this demonstration.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with MIL-T-3351 (for level A or B) and MIL-STD-1188 (for commercial) and specific requirements. Presence of one or more defects shall be cause for rejection.

## 5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each complete scraper, components, and attachments shall be preserved in accordance with level A preservation and packaging requirements of MIL-T-3351 except the cooling system shall be preserved as specified in 5.1.3.

5.1.2 Level B. Each complete scraper, components, and attachments shall be preserved in accordance with the level B preservation and packaging requirements of MIL-T-3351 except the cooling system shall be preserved as specified in 5.1.3.



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5.1.3 Commercial. Each complete scraper and all items issued with the scraper shall be suitably preserved and packed in accordance with MIL-STD-1188 for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrapping, pack compartmentization or cartonizing of the individual item. The cooling system shall be filled with a fresh mixed solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating in detail the preservation materials to be removed and any other servicing required prior to placing the scraper in operation.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete scraper, preserved as specified in 5.1, shall be packed in accordance with level A packing requirements of MIL-T-3351 as specified for type II, unboxed (mobile) shipment.

5.2.2 Level B. Each complete scraper, preserved as specified in 5.1, shall be packed in accordance with level B packing requirements of MIL-T-3351 as specified for type II, unboxed (mobile) shipment.

5.2.3 Commercial. Each complete scraper preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188. Commercial packing must protect items against physical and environmental damage during shipment, handling, and storage to assure safe delivery at destination.

5.3 Marking.

5.3.1 Military. Marking for level A or B protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The scraper covered by this specification is intended for use in general earthmoving work such as construction of military roads and airfields.



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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
(c) Time frame required for submission of the preproduction model (see 3.3).
(d) Technical publications required (see 3.30).
(e) Time frame for completing and submitting a technical information package (see 3.32).
(f) When items on the basic issue items list are required (see 3.33).
(g) Time frame for completing and submitting the vehicle classification data sheet (see 3.34).
(h) When winterized scrapers are required and number of scrapers to be furnished (see 3.35).
(i) Time frame required for submission of winterized scraper for examination and demonstration (see 3.35).
(j) When Swingfire heaters are required and number of heaters to be furnished (see 3.35.2).
(k) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production scrapers from the approved preproduction model during production will be subject to the approval of the Government. Approval of the preproduction model will not relieve the contractor of his obligation to furnish scrapers conforming to this specification.

6.4 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.29.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

Custodian:
Army - ME

User activity:
Army - AT

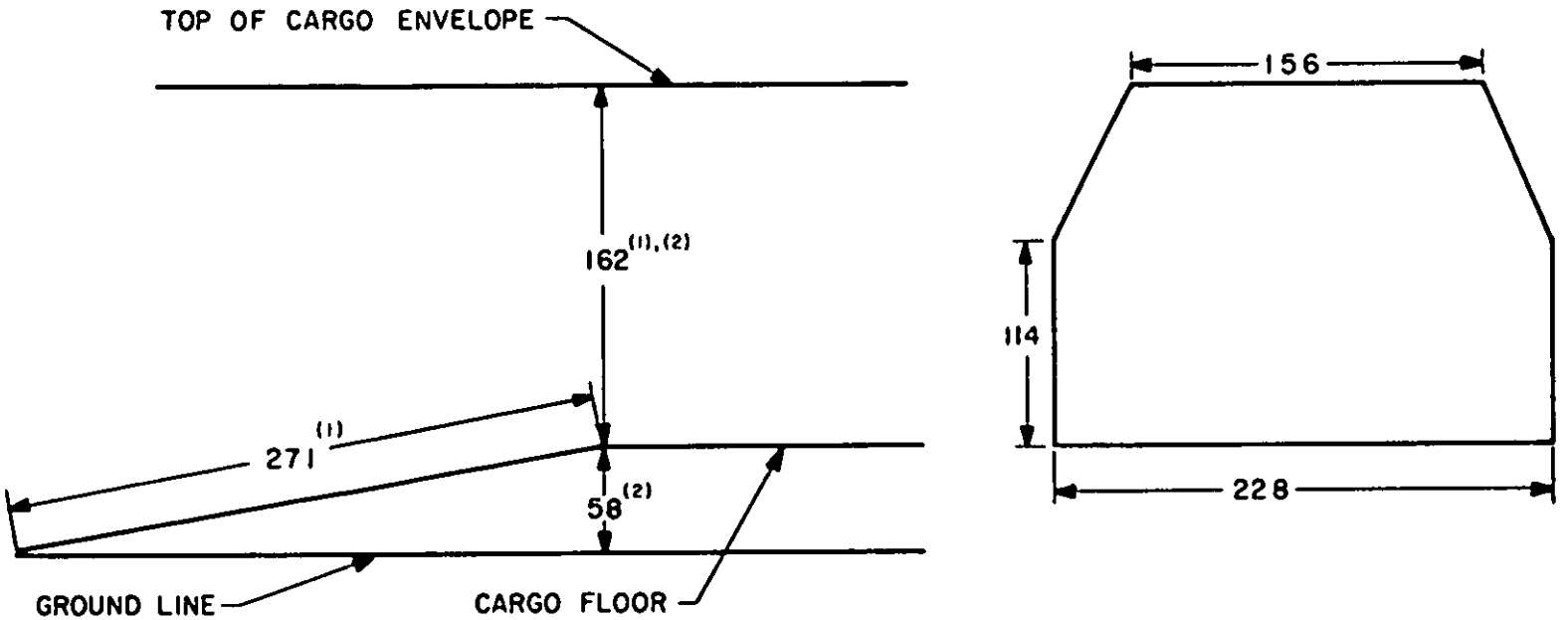
Preparing Activity:
Army - ME

Project 3805-A108



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FORWARD CARGO OPENING C-5A
FORWARD KNEEL POSITION




NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. REFERENCE: AMCP 706-130, ENGINEERING DESIGN HANDBOOK, DECEMBER 1967.
3. REFERENCE: USATECOM PROJECT NO 4-ES-065-050-004, SERVICE TEST OF C-5A AIRCRAFT, FIRST PARTIAL REPORT BY H.H JOHNSON, JR., CPT., JANUARY 1971.

FIGURE 1. C-5A. cargo envelope.

X-3171A



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| LOAD CLASS | | |
|---|---|---|
| E | C | H |
| | | |

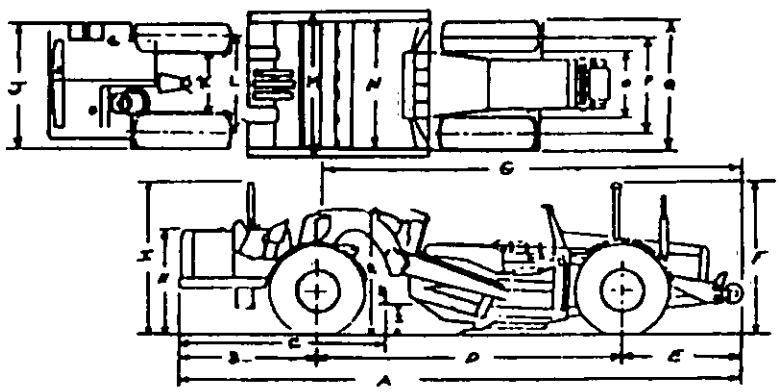
## VEHICLE CLASSIFICATION DATA
## TRACTOR SCRAPER

Data Required for Identification and Load Classification for Subject Vehicle

NOMENCLATURE:

| | |
|---|---|
| MFR. | MODEL NO. |
| FSN | Agency |

| Traveling Load Category | | Total Weight W | Pay Load |
|---|---|---|---|
| Empty | E | | |
| Cross Country or Off Highway | C | | |
| Laden or On Highway | H | | |




| Traveling Load Category | LOADS ON | | Front Tires | | | Rear Tires | | |
|---|---|---|---|---|---|---|---|---|
| | A | B | No/ axle | size | psi | No/ axle | size | psi |
| Empty | | | | | | | | |
| Cross Country or Off Highway | | | | | | | | |
| Laden or On Highway | | | | | | | | |

All loads in short tons.
All loads to include weight of crew and appurtenances.
Class Nos. for vehicle over 3,000 lb.
Longitudinal dimensions in feet.
Vertical and transverse dimensions in inches.

FIGURE 2. Vehicle classification data.

X-3776



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| | DIMENSION |
|---|---|
| A = Overall length of tractor-scraper. | A. _____ |
| B = ℄ of front (tractor) axle to front of tractor-scraper. | B. _____ |
| C = Length of tractor. | C. _____ |
| D = Wheelbase (tractor to scraper axle). | D. _____ |
| E = ℄ of rear (scraper) axle to rear of tractor-scraper. | E. _____ |
| F = Overall height (scraper). | F. _____ |
| G = ℄ of steering pivot to rear of tractor-scraper. | G. _____ |
| H = Height to top of tractor engine hood. | H. _____ |
| I = Height to top of tractor exhaust stack. | I. _____ |
| J = Width to outside of tractor tires. | J. _____ |
| K = Width to inside of tractor tires. | K. _____ |
| L = Tread width of tractor. | L. _____ |
| M = Overall width of tractor-scraper. | M. _____ |
| N = Width of cutting edge. | N. _____ |
| O = Width to inside of scraper tires. | O. _____ |
| P = Tread width of scraper. | P. _____ |
| Q = Width to outside of scraper tires. | Q. _____ |
| R = Height (overall) at gooseneck area with bowl fully raised. | R. _____ |
| R1 = Height (overall) at gooseneck area with bowl on ground. | R1. _____ |
| S = Ground clearance of tractor. | S. _____ |
| T = (T is not depicted in diagram.) Height at highest point of integral ROPS/FOPS cab. | T. _____ |

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Development Command
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-S-53000(ME) Scraper, 14-18 Cubic Yard Minimum Struck Capacity, Open Bowl, Single-DED, Commercial Construction Equipment (CCE)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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